

Assignment 13 (ELEC 341 L13_RLDesign)

Problem 1:

Given a unity feedback system that has the forward transfer function

$$G(s) = \frac{K(s-2)(s-4)}{(s^2+6s+25)} = K \frac{s^2-6s+8}{s^2+6s+25}$$

do the following:

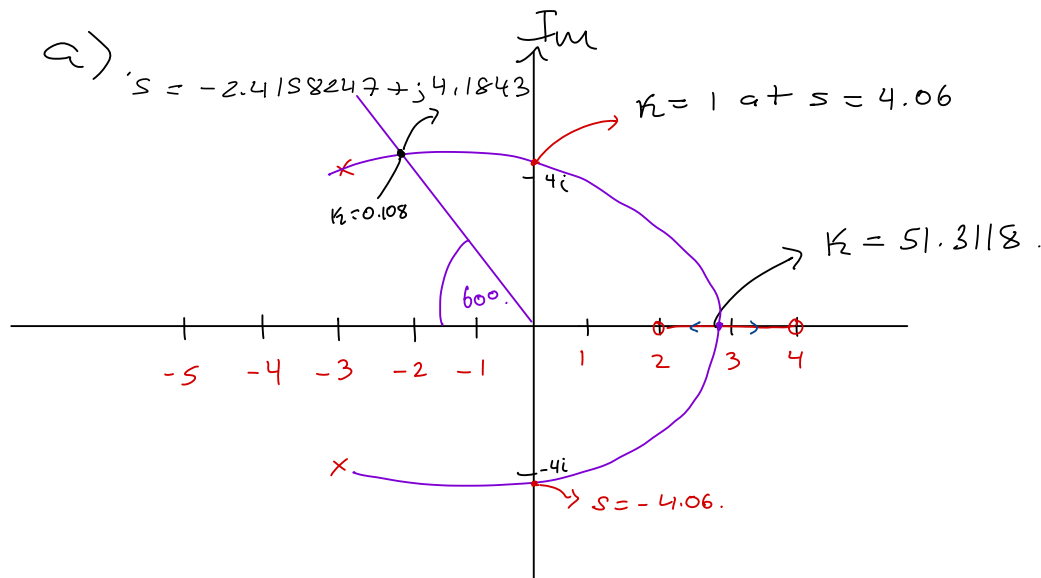
- Sketch the root locus.
- Find the imaginary-axis crossing.
- Find the gain, K , at the $j\omega$ -axis crossing.
- Find the break-in point.
- Find the point where the locus crosses the 0.5 damping ratio line.
- Find the gain at the point where the locus crosses the 0.5 damping ratio line.
- Find the range of gain, K , for which the system is stable.

$$s^2+6s+9+16$$

$$(s+3)^2 = -16$$

$$s = -3 \pm 4j$$

Please note that we do not expect an accurately drawn diagram, i.e., a rough sketch will suffice. You will only need to find and show the requested key points on the diagram and the rest can be drawn approximately.



$$b \& c) L(s) = \frac{s^2-6s+8}{s^2+6s+25}$$

$$1 + K L(s) = 0$$

$$s^2(6s+25) + K(s^2-6s+8) = 0$$

$$s^2 + 6s + 25 + Ks^2 - 6sK + 8K = 0$$

$$(1+K)s^2 + (6-6K)s + 25+8K = 0$$

$$s^2 \quad 1+K \quad 25+8K$$

$$s^1 \quad 6-6K \quad 0$$

$$s^0 \quad 25+8K$$

$$6-6K > 0$$

$$6 > 6K$$

$$K < 1$$

$$s^0 \quad K=1$$

1

jw axis crossings at $K=1$

$$P(s) \Rightarrow 2s^2 + 33 = 0,$$

$$2s^2 = -33$$

$$s = \pm j4.06202$$

$$d) \quad L(s) = \frac{N(s)}{D(s)} = \frac{s^2 - 6s + 8}{s^2 + 6s + 25}, \quad K = \frac{-1}{L(s)}$$

$$\frac{dL}{ds} \Rightarrow N'(s)D(s) - D'(s)N(s) = 0.$$

$$(2s-6)(s^2+6s+25) - (2s+6)(s^2-6s+8) = 0. \quad -6s^2 - 20s$$

$$\cancel{2s^3} + \cancel{12s^2} + 50s - \cancel{6s^3} - 36s - 150 - (\cancel{2s^3} - \cancel{12s^2} + 16s + \cancel{6s^3} - 36s + 48) = 0.$$

$$\cancel{2s^3} + \cancel{6s^2} + 14s - 150 - \cancel{2s^3} + \cancel{6s^2} + 26s - 48 = 0.$$

$$12s^2 + 34s - 198 = 0.$$

$$\boxed{s_1 = 2.88530316638} \text{ on RL.}$$

$$\underline{s_2 = -5.7186365}$$

$$L(s_1) = -1.948868$$

$$K = \frac{-1}{L(s_1)}$$

$$\boxed{K = 51.3118}$$

break in point
not on RL.

e & f

$$\gamma = 0.5$$

$$\gamma = \cos \theta$$

$$\cos \theta = 1/2$$

$$\theta = \pi/3 \text{ or } 60^\circ.$$

char eqn.

$$1 + K L(s) = 0.$$

$$1 + K \frac{s^2 - 6s + 8}{s^2 + 6s + 25} = 0.$$

$$s = -\gamma \omega_n + j \omega_n \sqrt{1 - \gamma^2}.$$

$$(1+K)s^2 - (6-6K)s + 25+8K = 0.$$

$$s^2 + \frac{6-6K}{1+K}s + \frac{25+8K}{1+K} = 0.$$

$$\omega_n^2 = \frac{25+8K}{1+K}$$

$$2\gamma\omega_n = \frac{6-6K}{1+K}$$

$$\omega_n^2 = \frac{25+8K}{1+K}$$

$$\omega_n^2 = \left(\frac{6-6K}{1+K}\right)^2$$

set equal.

$$\frac{25+8K}{1+K} = \left(\frac{6-6K}{1+K} \right)^2$$

$$(25+8K)(1+K) = (6-6K)^2$$

$$\cancel{25} + \cancel{8K} + \cancel{25K} + \cancel{8K^2} = 36 - 72K + 36K^2$$

$$33K \quad 0 = 28K^2 - 105K + 11$$

$$28K^2 - 105K + 11 = 0.$$

$$K_1 = 3.64213549 \text{ not pos cuz at jw axis } K=0.19$$

$$K_2 = 0.1078645$$

use this to determine ω_n .

$$K^2 + 36K,$$

and should increase towards break in point.

$$\omega_n = \frac{6-6K}{1+K}$$

$$\omega_n = 4.831649 \text{ rad/s}$$

$$s = -\gamma\omega_n \pm j\omega_n\sqrt{1-\gamma^2}, \quad \gamma = 0.5$$

$$s = -2.4158247 \pm j4.1843$$

$$g). (1+K)s^2 + (6-6K)s + 25+8K = 0.$$

$$s^2 \quad 1+K \quad 25+8K$$

$$s^1 \quad 6-6K \quad 0$$

$$s^0 \quad 25+8K$$

for stability

$$0 \leq K < 1$$

for stability.